

## NEWPORT SOUTH BEACH JETTY, OR, USA

WMO: 994280

Lat: 44.6130N

Lon: 124.0670W

Elev: 9

StdP: 101.22

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 0.0        | 1.4 | -8.6                          | 1.8 | 3.0  | -4.4 | 2.6 | 6.0  | 25.0                  | 9.5  | 21.3 | 10.5 | 4.5                   | 80   | 0.687 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.3                    | 18.5            | 14.4 | 17.3 | 14.5 | 16.4 | 14.4 | 16.0                | 16.9 | 15.5 | 16.3 | 15.0 | 15.8 | 4.4                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 15.7                            | 11.2 | 16.1 | 15.2 | 10.8 | 15.6 | 14.7 | 10.4 | 15.2 | 44.8          | 17.1 | 43.2 | 16.2 | 41.8 | 15.9 | 19.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |      |     |
| (4) 18.3          | 15.1 | 12.5 | DB                         | -2.1 | 26.6               | 2.0 | 4.1                                                | -3.5 | 29.6       | -4.6 | 32.0       | -5.8 | 34.3       | -7.2  | 37.2 | (4) |
| (5)               |      |      | WB                         | -4.3 | 17.9               | 2.7 | 1.2                                                | -6.2 | 18.8       | -7.8 | 19.5       | -9.3 | 20.1       | -11.3 | 21.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.3   | 8.2   | 7.8   | 8.2   | 8.9   | 10.6  | 12.0  | 12.7  | 13.0  | 12.9  | 11.3  | 9.6   | 7.7  |
|                                                                                  | DBStd       | 2.99   | 2.50  | 2.68  | 2.23  | 1.99  | 2.01  | 1.49  | 1.71  | 1.72  | 2.16  | 2.20  | 2.55  | 2.86 |
|                                                                                  | HDD10.0     | 387    | 66    | 70    | 64    | 43    | 15    | 1     | 1     | 0     | 1     | 11    | 36    | 80   |
|                                                                                  | HDD18.3     | 2951   | 315   | 296   | 314   | 284   | 241   | 191   | 174   | 165   | 163   | 217   | 261   | 331  |
|                                                                                  | CDD10.0     | 481    | 9     | 7     | 8     | 9     | 33    | 61    | 85    | 94    | 89    | 52    | 25    | 8    |
|                                                                                  | CDD18.3     | 2      | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 17     | 0     | 0     | 0     | 3     | 3     | 3     | 0     | 1     | 7     | 1     | 0     | 0    |
|                                                                                  | CDH26.7     | 4      | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 0     | 2     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 5.2    | 6.1   | 5.8   | 5.9   | 5.6   | 5.1   | 4.7   | 4.5   | 4.1   | 4.2   | 4.6   | 5.7   | 6.1  |
| Precipitation                                                                    | PrecAvg     | 1733   | 272   | 198   | 186   | 122   | 85    | 65    | 22    | 29    | 57    | 131   | 264   | 296  |
|                                                                                  | PrecMax     | 2613   | 540   | 427   | 332   | 243   | 183   | 158   | 82    | 177   | 159   | 269   | 516   | 570  |
|                                                                                  | PrecMin     | 1175   | 17    | 40    | 31    | 27    | 9     | 10    | 0     | 1     | 1     | 3     | 48    | 70   |
|                                                                                  | PrecStd     | 332    | 120   | 98    | 79    | 52    | 41    | 40    | 19    | 32    | 42    | 69    | 116   | 128  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 15.3  | 16.0  | 16.8  | 18.9  | 19.1  | 18.0  | 18.0  | 18.8  | 23.8  | 20.2  | 17.1  | 15.2 |
|                                                                                  |             | MCWB   | 9.9   | 9.1   | 10.1  | 11.6  | 13.2  | 13.0  | 16.2  | 15.1  | 13.5  | 12.0  | 13.3  | 12.3 |
|                                                                                  | 2%          | DB     | 13.4  | 13.6  | 13.5  | 13.6  | 15.3  | 15.7  | 16.9  | 17.4  | 18.3  | 16.6  | 14.9  | 13.3 |
|                                                                                  |             | MCWB   | 10.4  | 9.5   | 10.4  | 10.4  | 13.3  | 13.8  | 15.5  | 15.2  | 14.8  | 14.1  | 12.8  | 11.2 |
|                                                                                  | 5%          | DB     | 12.3  | 12.3  | 12.4  | 12.4  | 14.1  | 15.0  | 16.2  | 16.6  | 16.9  | 15.4  | 13.8  | 12.2 |
|                                                                                  |             | MCWB   | 10.6  | 9.3   | 10.4  | 10.3  | 12.3  | 13.4  | 15.0  | 15.0  | 14.6  | 13.8  | 12.5  | 10.6 |
|                                                                                  | 10%         | DB     | 11.6  | 11.2  | 11.5  | 11.7  | 13.2  | 14.3  | 15.4  | 15.7  | 15.9  | 14.4  | 12.9  | 11.3 |
|                                                                                  |             | MCWB   | 10.2  | 9.0   | 9.8   | 9.9   | 11.5  | 13.0  | 14.6  | 14.6  | 14.5  | 13.1  | 11.8  | 9.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.9  | 12.7  | 12.5  | 12.6  | 15.7  | 15.5  | 17.1  | 16.4  | 16.6  | 16.5  | 14.9  | 13.6 |
|                                                                                  |             | MCDB   | 13.2  | 13.3  | 13.2  | 14.6  | 17.2  | 15.8  | 17.7  | 17.1  | 18.1  | 16.7  | 15.7  | 14.3 |
|                                                                                  | 2%          | WB     | 11.8  | 11.1  | 11.6  | 11.7  | 14.0  | 14.6  | 15.7  | 15.7  | 15.8  | 15.2  | 13.9  | 12.2 |
|                                                                                  |             | MCDB   | 12.3  | 12.1  | 12.4  | 12.6  | 15.3  | 15.1  | 16.2  | 16.4  | 17.1  | 15.7  | 14.3  | 12.9 |
|                                                                                  | 5%          | WB     | 11.2  | 10.3  | 10.8  | 11.0  | 13.0  | 14.0  | 15.1  | 15.1  | 15.3  | 14.5  | 13.1  | 11.3 |
|                                                                                  |             | MCDB   | 11.8  | 11.3  | 11.6  | 11.9  | 14.1  | 14.6  | 15.6  | 15.8  | 16.3  | 15.1  | 13.6  | 11.9 |
|                                                                                  | 10%         | WB     | 10.5  | 9.6   | 10.2  | 10.4  | 12.3  | 13.4  | 14.3  | 14.5  | 14.9  | 13.8  | 12.4  | 10.4 |
|                                                                                  |             | MCDB   | 11.1  | 10.7  | 11.1  | 11.4  | 13.3  | 14.1  | 14.9  | 15.2  | 15.9  | 14.5  | 13.0  | 11.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.7   | 5.1   | 4.9   | 4.6   | 3.6   | 3.2   | 3.0   | 3.3   | 4.1   | 5.0   | 4.9   | 4.6  |
|                                                                                  |             | MCDBR  | 5.6   | 6.5   | 6.3   | 5.9   | 5.0   | 4.6   | 3.7   | 4.4   | 6.1   | 6.3   | 5.5   | 5.0  |
|                                                                                  |             | MCWBR  | 3.8   | 4.4   | 4.1   | 4.1   | 3.2   | 2.9   | 3.1   | 3.3   | 3.7   | 3.6   | 3.8   | 4.0  |
|                                                                                  | 5% WB       | MCDBR  | 3.9   | 4.8   | 4.6   | 4.9   | 4.5   | 3.6   | 3.5   | 4.0   | 4.7   | 4.0   | 4.0   | 4.2  |
|                                                                                  |             | MCWBR  | 3.4   | 3.9   | 4.0   | 3.8   | 3.2   | 2.7   | 2.9   | 3.2   | 3.4   | 3.2   | 3.7   | 3.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.325  | 0.324 | 0.336 | 0.370 | 0.377 | 0.362 | 0.349 | 0.356 | 0.348 | 0.336 | 0.331 | 0.331 |      |
|                                                                                  | taud        | 2.527  | 2.549 | 2.527 | 2.390 | 2.401 | 2.460 | 2.516 | 2.510 | 2.526 | 2.556 | 2.525 | 2.471 |      |
|                                                                                  | Ebn at Noon | 797    | 871   | 903   | 892   | 893   | 906   | 914   | 897   | 878   | 843   | 781   | 743   |      |
|                                                                                  | Edh at Noon | 68     | 78    | 91    | 113   | 115   | 109   | 102   | 99    | 90    | 77    | 67    | 66    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.32   | 2.15  | 3.19  | 4.57  | 5.69  | 6.18  | 6.68  | 5.65  | 4.44  | 2.69  | 1.51  | 1.09  |      |
|                                                                                  | RadStd      | 0.19   | 0.24  | 0.35  | 0.34  | 0.47  | 0.46  | 0.45  | 0.44  | 0.35  | 0.28  | 0.19  | 0.15  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (24 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +17     |

Nomenclature: See separate page